

# Homework 1

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## Exercise 1

Probabilities are sensitive to the form of the question that was used to generate the answer

My neighbor has two children. Assuming that the gender of a child is like a coin flip, it is most likely, a priori, that my neighbor has one boy and one girl, with probability  $1/2$ . The other possibilities-two boys or two girls-have probabilities  $1/4$  and  $1/4$ .

a. Suppose I ask him whether he has any boys, and he says yes. What is the probability that one child is a girl?

b. Suppose instead that I happen to see one of his children run by, and it is a boy. What is the probability that the other child is a girl?

## Exercise 2

Deriving the inverse gamma density

Let  $X \sim \text{Ga}(a, b)$ , i.e.

$$\text{Ga}(x \mid a, b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-xb}$$

Let  $Y = 1/X$ . Show that  $Y \sim \text{IG}(a, b)$ , i.e.,

$$\text{IG}(x \mid \text{shape} = a, \text{scale} = b) = \frac{b^a}{\Gamma(a)} x^{-(a+1)} e^{-b/x}$$

Hint: use the change of variables formula.

## Exercise 3

MLE minimizes KL divergence to the empirical distribution Let  $p_{\text{emp}}(x)$  be the empirical distribution, and let  $q(x \mid \theta)$  be some model. Show that  $\text{argmin}_q D_{\text{KL}}(p_{\text{emp}} \parallel q)$  is obtained by  $q(x) = q(x; \hat{\theta})$ , where  $\hat{\theta}$  is the MLE.

Hint: use non-negativity of the KL divergence.

## Exercise 4

MLE for the uniform distribution

Consider a uniform distribution centered on 0 with width  $2a$ . The density function is given by

$$p(x) = \frac{1}{2a} I(x \in [-a, a])$$

a. Given a data set  $x_1, \dots, x_n$ , what is the maximum likelihood estimate of  $a$  (call it  $\hat{a}$ )?

b. What probability would the model assign to a new data point  $x_{n+1}$  using  $\hat{a}$ ?

c. Do you see any problem with the above approach? Briefly suggest (in words) a better approach.

## Exercise 5

Uncorrelated does not imply independent

Let  $X \sim U(-1, 1)$  and  $Y = X^2$ . Clearly  $Y$  is dependent on  $X$  (in fact,  $Y$  is uniquely determined by  $X$ ). However, show that  $\rho(X, Y) = 0$ .

Hint: if  $X \sim U(a, b)$  then  $E[X] = (a + b)/2$  and  $\text{var}[X] = (b - a)^2/12$

## Exercise 6

Uncorrelated and Gaussian does not imply independent unless jointly Gaussian

Let  $X \sim \mathcal{N}(0, 1)$  and  $Y = WX$ , where  $p(W = -1) = p(W = 1) = 0.5$ . It is clear that  $X$  and  $Y$  are not independent, since  $Y$  is a function of  $X$ .

a. Show  $Y \sim \mathcal{N}(0, 1)$ .

b. Show  $\text{cov}[X, Y] = 0$ . Thus  $X$  and  $Y$  are uncorrelated but dependent, even though they are Gaussian.

Hint: use the definition of covariance

$$\text{cov}[X, Y] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

and the rule of iterated expectation

$$\mathbb{E}[XY] = \mathbb{E}[\mathbb{E}[XY | W]]$$

## Exercise 7

Bivariate Gaussian

Let  $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  where  $\mathbf{x} \in \mathbb{R}^2$  and

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

where  $\rho$  is the correlation coefficient. Show that the pdf is given by

$$p(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[ \frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} - 2\rho \frac{(x_1 - \mu_1)}{\sigma_1} \frac{(x_2 - \mu_2)}{\sigma_2} \right]\right)$$

## Exercise 8

### Conditioning a bivariate Gaussian

Consider a bivariate Gaussian distribution  $p(x_1, x_2) = \mathcal{N}(x \mid \mu, \Sigma)$  where

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix} = \sigma_1 \sigma_2 \begin{pmatrix} \frac{\sigma_1}{\sigma_2} & \rho \\ \rho & \frac{\sigma_2}{\sigma_1} \end{pmatrix}$$

where the correlation coefficient is given by

$$\rho \triangleq \frac{\sigma_{12}}{\sigma_1 \sigma_2}$$

- a. What is  $P(X_2 \mid x_1)$ ? Simplify your answer by expressing it in terms of  $\rho, \sigma_2, \sigma_1, \mu_1, \mu_2$  and  $x_1$ .
- b. Assume  $\sigma_1 = \sigma_2 = 1$ . What is  $P(X_2 \mid x_1)$  now?

### Exercise 9

Sensor fusion with known variances in 1d

Suppose we have two sensors with known (and different) variances  $v_1$  and  $v_2$ , but unknown (and the same) mean  $\mu$ . Suppose we observe  $n_1$  observations  $y_i^{(1)} \sim \mathcal{N}(\mu, v_1)$  from the first sensor and  $n_2$  observations  $y_i^{(2)} \sim \mathcal{N}(\mu, v_2)$  from the second sensor. (For example, suppose  $\mu$  is the true temperature outside, and sensor 1 is a precise (low variance) digital thermosensing device, and sensor 2 is an imprecise (high variance) mercury thermometer.) Let  $\mathcal{D}$  represent all the data from both sensors. What is the posterior  $p(\mu \mid \mathcal{D})$ , assuming a non-informative prior for  $\mu$  (which we can simulate using a Gaussian with a precision of 0)? Give an explicit expression for the posterior mean and variance.